

Andrew Yuan

📍 New York ✉ aycooper2022@gmail.com 🌐 My Website 📄 GitHub

Education

Harvard University

August 2026 - Present

M.Eng. Computational Science and Engineering

The Cooper Union for the Advancement of Science and Art

August 2022 - May 2026

B.Eng. in Electrical Engineering, summa cum laude

Minors in Mathematics and Computer Science

GPA: 3.94/4.00 | Class Rank: 1/30

- **Awards/Honors:** Abraham Pletman Memorial Fund Prize (Highest GPA in Electrical Engineering Class of 2026), Howard Flagg EE'75 Memorial Prize, Jesse Sherman EE'33 Book Award, Dean's List (All Semesters)
- **Relevant Graduate Coursework:** Numerical Analysis, Bayesian Machine Learning, Convex Optimization Techniques, Medical Imaging, Big Data for Finance, Financial Signal Processing

Research Experience

Large Language Model-based Psychotherapy Evaluations

New York, NY

Advisor: Dr. Xiaofan Jiang, Columbia University

November 2025 – June 2026

- Conducted a literature review on the use of large language models (LLMs) and multimodal LLMs for psychotherapy and identified a gap in multi-turn behavioral analysis.
- Designed an experimental framework with standardized mental health role-playing prompts to simulate multi-turn psychotherapy, utilizing RoBERTa-based transformer models to perform sentiment classification on conversational data and analyzing emotion vectors to quantify LLM response behavior.

NSF REU: Kernel Engineering in Gaussian Processes

Pomona, CA

Advisor: Dr. Jimmy Risk, Cal Poly Pomona

June 2025 – August 2025

- Developed an experimental framework with automated training loops and batched evaluations using BoTorch and GPyTorch to benchmark Gaussian process (GP) kernels for high-dimensional Bayesian optimization (BO).
- Logged and stored experimental outputs (e.g. model predictions, hyperparameters) in structured CSV and pickle formats for analysis and visualization using Matplotlib.
- Extended results using reproducing kernel Hilbert space (RKHS) theory, constructing counterexamples showing linear regret in BO under model misspecification and underestimated RKHS norm bounds.

Professional Experience

Image Processing Intern

Santa Clara, CA

OmniVision Technologies

May 2026 – Present

- Developed Python image analysis tools for medical camera applications, supporting color detection, spectral analysis, and quantitative scoring across multiple color-space and gamut representations.
- Refactored and expanded legacy imaging code for integration with new camera simulation software, implementing additional demosaicing methods and support for color filter array patterns.

Projects

Compressed Sensing MRI Reconstruction

[Link](#)

- Implemented Fourier and wavelet-domain compression experiments on T1 and T2-weighted MRI data, comparing Haar, Daubechies-4, and Coiflet-3 bases across multiple decomposition levels.
- Designed a compressed sensing reconstruction experiment utilizing wavelet sparsity regularization and gradient descent, analyzing the tradeoff between sampling rate, image quality, and compression.

Spectral-Domain OCT Reconstruction

[Link](#)

- Reconstructed spatial-domain OCT A-scans and B-scan images from raw spectral OCT data using background subtraction, deconvolution, Hamming windowing, k-domain resampling, and FFT processing.
- Applied spectral-domain phase microscopy to complex A-scan time series to estimate nanometer-scale speaker vibrations and recover frequency spectra, identifying an 8 kHz single-tone response and 40 multi-tone components.

Transformer-TimeGAN for Financial Time-Series

[Link](#)

- Modified TimeGAN architecture with Transformer-based embedder and discriminator components to evaluate

self-attention mechanisms for synthetic financial time-series generation.

- Improved synthetic stock generation performance by replacing the GRU-based discriminator with a Transformer discriminator, improving discriminative score by 33.8% and predictive score by 4.0% relative to baseline TimeGAN.

Multi-Purpose Analog Hall-Effect Keypad

[Link](#) 

- **Awarded 2026 Best Senior Electrical Engineering Project**
- Designed and prototyped an open-source analog Hall-effect keypad using 3D-printed Cherry MX-style switches, AH49E sensors, MCP3008 ADC, and a Pico 2 microcontroller to capture continuous key-displacement data.
- Developed embedded firmware in C and Python host software for 1 kHz sensor acquisition, USB HID reporting, real-time ADC visualization, and configurable threshold-based mapping of key displacement to user-defined outputs.

Teaching and Mentorship

Math Department Tutor and Grader

August 2024 - May 2026

The Cooper Union

- Hosted individual one-on-one and weekly drop-in tutoring sessions for 5+ students in: Calculus I-II, Vector Calculus, Linear Algebra, Probability, Differential Equations, and Discrete Mathematics.
- Evaluated and provided feedback for 60+ students in Calculus I-II and Intro to Linear Algebra courses, meeting weekly with course instructor to review student performance.

IEEE Cooper Union Mentor

August 2024 - May 2026

The Cooper Union

- Mentored first-year Electrical Engineering and Computer Science students, providing tutoring for Electrical and Computer Engineering courses, and advising on course planning and professional development.

Leadership Experience

International Finance Student Association Chapter Co-Founder

August 2024 - May 2026

The Cooper Union

- Co-founded the first North American chapter of International Finance Student Association at Cooper Union, growing membership to 30+ active participants, surpassing the average club size of 10.
- Organized guest-speaker events and workshops with a budget of \$1000 to promote professional development and provide networking opportunities.

Assistant Badminton Coach

May 2022 - June 2026

New Jersey Badminton Club

- Designed structured training programs for 30+ junior and adult athletes in weekly group sessions, supplementing students with individual private training to deliver targeted technical feedback.
- Mentored and developed students to the US Junior National Team and Top 10 national junior rankings.

Cooper Union Badminton Club Liason

August 2024 - May 2026

The Cooper Union

- Organized biweekly intramurals (15+ participants) in coordination with local badminton clubs and two annual Cooper Union badminton tournaments with 20+ competitors each.

Eta Kappa Nu Delta Chi Chapter President

August 2025 - May 2026

The Cooper Union

- Supported chapter operations and member engagement, including planning initiation ceremony for 7 new members.

Athletics

Professional Badminton Athlete

2012-Present

USA Badminton Association

- **US National Team:** FISU World University Games (2025), BWF World Junior Championships (2022), Pan Am Junior Championships (2014-2022)
- Junior National Rank #1, Adult National Runner-Up, Junior National Champion, 2x Pan Am Junior Champion

Skills

- **Languages:** Python, C, C++, MATLAB, SQL, Java, HTML, $\text{\LaTeX}$
- **Libraries:** NumPy, Pandas, SciPy, scikit-learn, scikit-image, PyTorch, GPyTorch, OpenCV, Matplotlib
- **Tools:** Linux, Git, Bash, Docker, PostgreSQL, Jupyter Notebook, Spyder, VS Code, OpenSCAD, LTspice